INFORMATION TO USERS

The most advanced technology has been used to photograph and reproduce this manuscript from the microfilm master. UMI films the text directly from the original or copy submitted. Thus, some thesis and dissertation copies are in typewriter face, while others may be from any type of computer printer.

The quality of this reproduction is dependent upon the quality of the copy submitted. Broken or indistinct print, colored or poor quality illustrations and photographs, print bleedthrough, substandard margins, and improper alignment can adversely affect reproduction.

In the unlikely event that the author did not send UMI a complete manuscript and there are missing pages, these will be noted. Also, if unauthorized copyright material had to be removed, a note will indicate the deletion.

Oversize materials (e.g., maps, drawings, charts) are reproduced by sectioning the original, beginning at the upper left-hand corner and continuing from left to right in equal sections with small overlaps. Each original is also photographed in one exposure and is included in reduced form at the back of the book.

Photographs included in the original manuscript have been reproduced xerographically in this copy. Higher quality 6" x 9" black and white photographic prints are available for any photographs or illustrations appearing in this copy for an additional charge. Contact UMI directly to order.

PREVIEW

U·M·I

University Microfilms International
A Bell & Howell Information Company
300 North Zeeb Road Ann Arbor MI 48106-1346 USA
313 761-4700 800 521-0600

PREVIEW

Order Number 9034910

Universal cycles: On beyond de Bruijn

Hurlbert, Glenn Howland, Ph.D.

Rutgers The State University of New Jersey - New Brunswick, 1990

PREVIEW

Copyright ©1990 by Hurlbert, Glenn Howland. All rights reserved.

U·M·I
300 N. Zeeb Rd.
Ann Arbor, MI 48106

PREVIEW

UNIVERSAL CYCLES: ON BEYOND DE BRUIJN

BY GLENN HOWLAND HURLBERT

A dissertation submitted to the

Graduate School - New Brunswick

Rutgers, The State University of New Jersey

in partial fulfillment of requirements

for the degree of

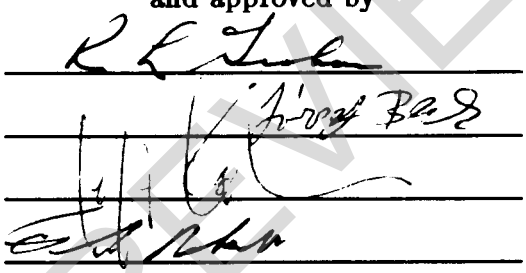
Doctor of Philosophy

Graduate Program in Mathematics

Written under the direction of

Professor Ronald L. Graham

and approved by



New Brunswick, New Jersey

May, 1990

PREVIEW

© 1990

Glenn Howland Hurlbert

ALL RIGHTS RESERVED

PREVIEW

ABSTRACT OF THE DISSERTATION

Universal Cycles: On Beyond de Bruijn

by Glenn Howland Hurlbert

Dissertation Director

Professor Ronald L. Graham

Universal Cycles have their origin in the 1940's with the discovery of de Bruijn Cycles, the most compact way of listing all binary n-tuples. A de Bruijn Cycle is a cyclic listing of 0's and 1's of length 2^n with the property that every binary n-tuple appears exactly once consecutively on the cycle. More recently Chung, Diaconis, and Graham have inaugurated a more general study of compact listings (of the same cyclical type) of other combinatorial structures. These include k-ary n-tuples, permutations and k-permutations of $[n] = \{1, 2, \ldots, n\}$, partitions and k-partitions of $[n]$, k-subsets of $[n]$, and k-dimensional subspaces of an n-dimensional vector space over a finite field, among others. Even higher dimensional versions are considered (de Bruijn tori, etc.). This dissertation serves to present the reader with some of the history of the subject as well as with the most recent results and progress in the area. In addition, we consider the question of the Hamiltonicity of the symmetric bipartite graph induced by the subsets of $[n]$ of size k and $(n-k)$ with particular emphasis on the case $n = 2k+1$. This question is commonly known as the Antipodal Layers Problem, the special case known as the Middle Layers Problem.

PREVIEW

ACKNOWLEDGEMENTS

I would like to express my appreciation to many of the professors who have inspired me in the past. In particular, thanks to Elmer Hayashi and Fred Howard, who put up with me at Wake Forest University, to Joel Spencer, who at Stony Brook introduced me to combinatorics and to Paul Erdős, who seems to inspire all those he meets, and to Ron Graham, who introduced me to Universal Cycles while at Rutgers and under whose guidance I have learned a great deal. Thanks to Jeff Kahn, Fred Roberts, and Józef Beck for many helpful comments and for serving on my defense committee. Thanks also to my sichadic roommate Robert Hochberg, whose mathematical and Christian friendship I will never forget. Finally, I must thank my parents, whose constant love and support I could not do without.

PREVIEW

DEDICATION

This dissertation is dedicated with love to the memories of my grandfather, Luther Carrington Goodrich, and uncle, Frank Chauncey Goodrich.

PREVIEW

TABLE OF CONTENTS

	page
Title	i
Abstract	ii
Acknowledgements	iii
Dedication	iv
Table of Contents	v
List of Tables and Figures	vii
Chapter One — Introduction	1
Chapter Two — de Bruijn Cycles	3
2.1 An Application	3
2.2 Existence Proof	5
2.3 Symmetric Cycles	8
2.4 Algorithms	9
2.5 de Bruijn Tori	12
2.6 Bibliography	13
Chapter Three — Universal Cycles	15
3.1 Permutations of $[n]$	15
3.1.1 k-Permutations	15
3.1.2 Order-isomorphism	16
3.2 Partitions and k-Partitions of $[n]$	31
3.3 k-Subsets of $[n]$	39

PREVIEW

3.3.1 A Necessary Condition 39
3.3.2 $k = 2$ 41
3.3.3 General k 43
3.3.4 Multi-Covers 51
3.4 Vector Spaces 58
3.5 Bibliography 61

Chapter Four — The Antipodal Layers Problem 62
4.1 Introduction 62
4.2 Middle Layers 62
4.3 Antipodal Layers 67
4.4 Remarks 76
4.5 Bibliography 79

Vita 80

PREVIEW

LIST OF TABLES AND FIGURES

		page
2.1	de Bruijn graphs $dB(2,2)$ and $dB(3,2)$.	6
2.2	The "preferred ones" tree in $dB(3,2)$.	10
2.3	The square (2,2,2)-de Bruijn Torus.	12
2.4	A (2,2,3)-de Bruijn Torus with $S_1 = 0011$.	13
3.1	Order-isomorphic representation.	16
3.2	Displacement for $n = 5$.	17
3.3	$\mathcal{T}_3$.	19
3.4	Relations from a UCycle for $n = 3$.	19
3.5	Poset $\mathcal{P}$ for $n = 3$ with labellings.	20
3.6a	Permutation list for $\mathcal{K}_3$.	22
3.6b	Temporary list for $\mathcal{K}_4$.	22
3.7	Keys for $\mathcal{K}_5$.	23
3.8	The "breaker" permutation $p_0 \in \mathcal{K}_6$.	25
3.9a	Key UCycle $\mathcal{K}_4$.	25
3.9b	Key UCycle $\mathcal{K}_5$.	25
3.10	Displacement drops last value by 2.	27
3.11	The tail end of $\mathcal{K}_7$.	28
3.12	Breaker sequence rooted at $(n-1\ldots1)$ in T_n.	29
3.13	$\mathcal{T}_4$ is $\mathcal{P}_3$ with 2 multiple edges.	33
3.14	Breaker sequences in $\mathcal{T}_n$.	34
3.15	Hamiltonian cycle in $\mathcal{P}_{n-1,n}$.	36
3.16	UCycle for 6-partitions of [7].	36
3.17	UCycle construction for $k = 2$.	42

PREVIEW

3.18	Connectedness of $\mathcal{H}_{n,3}$.	48
3.19	Connectedness of $\mathcal{H}_{n,4}$.	48
3.20	Connectedness of $\mathcal{H}_{n,6}$.	50
3.21	Connectedness of $\mathcal{H}'_n$.	56
4.1	$\mathcal{M}_3 \cong \mathcal{Q}^3 \bowtie \mathcal{Q}^4$.	64
4.2	$\mathcal{M}_3 \cong \mathcal{Q}^2 \bowtie \mathcal{N}_3(5)$.	65
4.3	Combining "half-cycles".	69
4.4	Cases for small k not covered by Theorem 4.2.	70
4.5	Representatives of a cycle in $\mathcal{D}$.	72
4.6	The order of towns for the case $k = 5$.	76

PREVIEW

Chapter 1
Introduction

The following sequence of zeros and ones has a rather curious property.

00011101

If each of the eight digits is taken to be the starting point of a triple to be listed, and the end of the sequence is thought of as looping back to the beginning, as if on a circle, then this particular list would include every binary triple exactly once. Indeed, 000, 001, 011, 111, 110, 101, 010, and 110 is precisely this list. It so happens that there are exactly 2 such sequences with this property (up to cyclic permutations), the other being the reverse ordering of these same digits.

Such sequences have come to be known as *de Bruijn Cycles*, this example being one which cycles through binary triples. More generally, we may wish to cycle through binary n-tuples, or even k-ary n-tuples for positive integers k and n. For clarity, this latter case would be a cyclic listing of k^n integers from the set $\{0, 1, \ldots, k-1\}$ such that every n-tuple of such integers appears exactly once consecutively on the cycle. Questions one might ask would include those of existence, construction, and enumeration, among others. Chapter 2 addresses these questions and includes interesting proofs and algorithms, together with applications and further generalizations.

Chapter 3 includes a full discussion of *Universal Cycles*, in short *UCycles*, which are a generalization of de Bruijn Cycles due to F.R.K.Chung, P.Diaconis, and R.L.Graham and can be described as follows. Given a family $\mathcal{F}$ of objects

PREVIEW

sharing some combinatorial structure of "rank" k, one constructs a cyclic sequence of length $|\mathcal{F}|$, using elements from some ground set $\mathcal{X}$, so that each member of $\mathcal{F}$ is represented exactly once as some k consecutive symbols on the cycle. For example, with de Bruijn Cycles, a binary n-tuple is represented by the actual n-tuple itself, whereas a k-dimensional subspace of an n-dimensional vector space over a finite field may be represented by any k vectors which span the subspace. Structures considered in this dissertation are permutations of n objects, partitions of $[n]$, k-subsets of $[n] = \{1, 2, \ldots, n\}$, and k-dimensional subspaces of an n-dimensional vector space over a finite field. Similar questions are raised regarding each of the structures and, additionally, we address this notion of representation, an unnecessary notion in the de Bruijn case. As it turns out, the topics become more difficult as one proceeds along our list of structures. Those we have yet to consider include finite projective planes, Steiner Triple Systems, and other designs. One may, in fact, attempt UCyles for trees or other classes of graphs or matroids. In all of these cases the key is always to find a nice way to represent any given member by some k-tuple.

Finally, Chapter 4 considers a certain problem regarding Hamiltonian cycles in graphs. The Antipodal Layers Problem deals with a large class of vertex transitive bipartite graphs induced by certain elements of the n-cube (the set of binary n-tuples). For $k < n/2$ one defines the graph $\mathcal{M}_{n,k}$ to have vertices all subsets of $[n]$ of size k or $n - k$ and edges between vertices for which one contains the other. It is conjectured that these graphs are Hamiltonian for all k and $n > 2k$, and here we prove that this is so for $n > ck^2$ for some constant c. It is possible that a Hamiltonian cycle in this graph is equivalent in some cases to a UCycle for k-subsets of $[n]$, though this appears difficult to resolve. The Antipodal Layers Problem is part of the larger question of the Hamiltonicity of vertex transitive graphs in general.

PREVIEW

Chapter 2
de Bruijn Cycles

2.1 An Application

To repeat, a *k-ary de Bruijn Cycle of order n* is a cyclic listing of length k^n of digits from the alphabet $\{0, 1, \ldots, k-1\}$ with the property that every n-tuple in this alphabet appears exactly once consecutively on the cycle. In 1894, Flye-Sainte Marie (see [6]) first discovered that such cycles exist for all n and k, in response to the question by Riviere. This work went largely unnoticed until the cycles were rediscovered in 1946 by de Bruijn, who was able to count the number of such cycles for the case $k = 2$ (see [3]). In 1951 van Aardenne-Ehrenfest and de Bruijn [1] were able to show that the number $b(n, k)$ of such cycles for arbitrary k is given by the formula

$$b(n, k) = (k!)^{k^{n-1}}/k^n.$$

These cycles have found uses in coding and communications (see [7]) as period generators and as random bit and random word generators, and there is even an interesting application to the theory of numbers, an application we now describe.

When defining normal numbers we consider only the digits following the decimal place, i.e. we take all numbers modulo 1. An infinite sequence of digits in base k is said to be *random* if, for all n, each n-tuple of digits occurs consecutively with equal asymptotic density k^{-n}. That is, if B is any block of n digits and $m_j(B)$ is the number of times it appears consecutively in the first j places, then $m_j(B) \to k^{-n}$ as $j \to \infty$. A number is then *normal to base k* if its digits

PREVIEW

form a random sequence. Alternatively, x is normal to base k if the numbers x, kx, k^2x, ..., are uniformly distributed modulo 1 (see [14]). It is well known that almost all real numbers are normal to a given base, but constructions are rare. In fact, almost all real numbers are normal to every base, but no such number has been found.

Here we use de Bruijn Cycles to construct uncountably many normal numbers in a fixed base k. Of course, once you find one such number, you may delete and/or insert any finite number of digits to obtain only countably many others, but that's cheating anyway. We begin with the realization that, given any infinite sequence b_1, b_2, ..., we can construct an isomorphism between the set of strings $x_1x_2\ldots$, where each x_i is a nonnegative integer less than b_i, and the interval $[0,1)$, given by

$$x = \sum_{n=1}^{\infty} x_n/p_n,$$

where $p_n = \prod b_{i\leq n}$. For our purposes, $b_n = kb(n,k)$.

Notice that a k-ary de Bruijn Cycle of order n contains every t-tuple k^{n-t} times as a consecutive block. In particular, for $t \leq n$ all t-tuples occur with equal frequency. Having revealed the secret the construction is obvious. We merely juxtapose sequences C_n of increasing orders $n = 1, 2, \ldots$, taking care that the first $n-1$ digits of C_n match those of C_{n-1}, which can happen in k ways. Normality and uncountability then follow. Unfortunately, it is easy to prove that, as with Cantor's set, this set has Lebesgue measure zero. Indeed, let μ be the measure of all normal numbers constructed in this manner. Then by prefacing these numbers with any t digits we construct another set of normal numbers whose measure is μ/k^t. The union over all possible t digits is then a set of measure μ, and the union over all t has infinite measure unless $\mu = 0$. Applications to random generators should be clear now, since any C_n approximates randomness for any k as n gets large. The reader is referred to [8], [9], and [17] for further discussion in this area.

PREVIEW

2.2 Existence Proof

The example first presented with $(n,k) = (3,2)$ was constructed using the *preferred ones* algorithm, easily described as placing an initial n zeros followed sequentially by ones "whenever possible". That is, write a 1 if the resulting n-tuple has not yet been listed, otherwise write a 0. We can use *preferred zeros* by reversing the natural order of 0 and 1. More generally, a preferential algorithm for arbitrary k first orders the alphabet $\{0,1,\ldots,k-1\}$ (naturally or otherwise), starts with n copies of the least element, and sequentially places the greatest element possible, avoiding duplication. For example, with $(n,k) = (2,3)$ and the ordering $1 < 2 < 0$, we construct the ternary de Bruijn Cycle for pairs

110020122.

Why these algorithms work is definitely not obvious, but put in a more general setting things become clearer.

The analysis of de Bruijn Cycles is best done in terms of graph theory, the setting first used by de Bruijn (see [3]). What we want is a Hamiltonian cycle (i.e. a path along edges visiting each vertex exactly once while returning to the original vertex) in a graph whose vertices represent all n-tuples. It remains to decide what edges are present, but this is an easy task. Call the first $(n-1)$ positions of an n-tuple its *prefix*, the last $(n-1)$ its *suffix*. By noticing that two consecutive n-tuples a and b on a de Bruijn Cycle must share $n-1$ positions, we realize this graph must have a directed edge from a to b whenever the suffix of a matches the prefix of b. This, then, is a kind of "transition" graph, telling us the possible ways to pass from one stage to the next. We call this the de Bruijn graph, $dB(n,k)$. Figure 2.1 shows the two graphs $dB(2,2)$ and $dB(3,2)$.

Notice that the edges of $dB(2,2)$ are labelled in the obvious way by the vertices of $dB(3,2)$. This tells us that, rather than looking for a Hamiltonian cycle in

PREVIEW

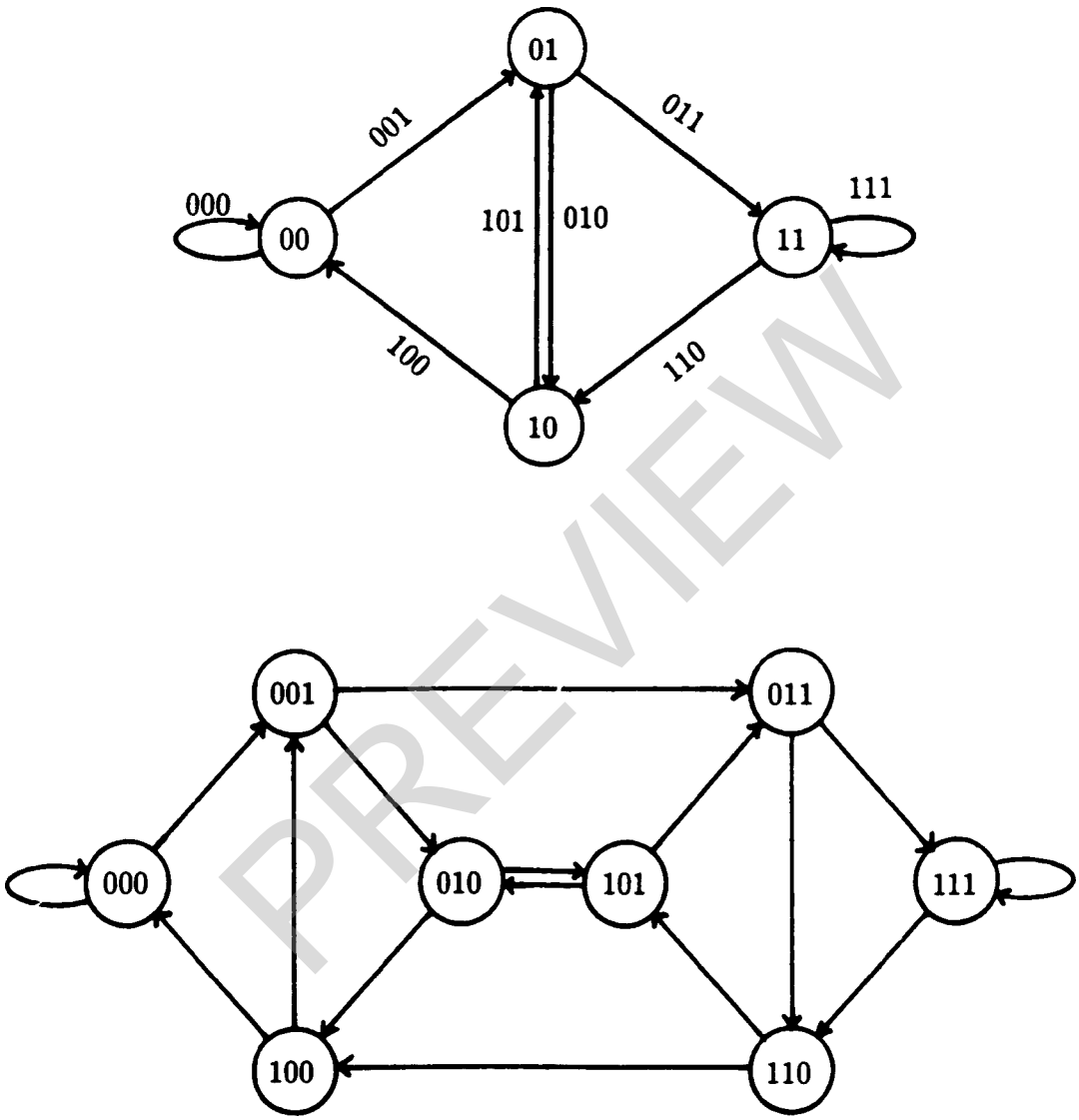


Figure 2.1. $dB(2,2)$ (top) and $dB(3,2)$ (bottom).

PREVIEW

Reproduced with permission of copyright owner. Further reproduction prohibited without permission.